VIETNAMESE - GERMAN UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



Customer Segmentation Analysis REPORT

Customer and Segmentation Data Analytics

Instructor: Phan Trọng Nhân, PhD

Student 1: Dương Quý Trang - 10423110
Student 2: Nguyễn Hà Khải - 10423053
Student 3: Nguyễn Quang Ngọc Trác - 10423109
Student 4: Tô Huỳnh Phúc - 10423093
Student 5: Nguyễn Hữu Triển - 17813

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Chapter 1

Introduction

1.1 Context

In retail and e-commerce enterprise systems, understanding customer purchasing patterns is vital for sustainable growth. Analysis of initial transactional data at CompanyX revealed four core business pain points:

- **High Single-Purchase Churn:** 62.9% of customers make only one order and never return. Only 29.5% make a second order, and a minimal 7.4% complete 3 to 5 orders.
- **Basket Size Limitations:** A significant portion of transactions involve minimal item counts (2,492 single-item purchases vs. 4,480 two-item purchases), indicating low initial basket size expansion.
- **Cross-Selling Inefficiencies:** Over 9,000 B2C customers purchased products in the "Bikes" category but never bought complementary items from "Accessories" or "Clothing".
- **Customer Inactivity:** A high proportion of customer profiles remain inactive for extended durations ($>$ 180 days), reducing total Customer Lifetime Value (CLV) and inflating Customer Acquisition Costs (CPA).

Chapter 2

Theoretical Framework & Industry Benchmark Case Studies

2.1 Case Study Summary

To formulate an effective solution for CompanyX, three industry benchmarks utilizing distinct segmentation methodologies were evaluated:

2.1.1 Black Diamond (RFM Segmentation)

Black Diamond, specializing in high-end technical climbing and outdoor equipment, transitioned from wholesale to Direct-to-Consumer (D2C) online [**Lexer2025**]. Faced with messy data across disparate systems, mass email marketing, and high customer churn, they established a Single Customer View (SCV) and applied Recency, Frequency, and Monetary (RFM) segmentation. Personalized email campaigns targeting inactive (> 180 days) and lapsed (> 1 year) customers yielded a +1,101% revenue per email increase, a 50% reduction in CPA, and a 2x Return on Advertising Spend (ROAS).

2.1.2 Zappos (K-Means Clustering)

Zappos is an e-commerce platform for footwear and apparel [**TowardsAI2020**]. Facing uniform marketing inefficiencies, they deployed unsupervised K-Means clustering on Annual Income and Spending Score data. Using the Elbow Method, they identified 5 distinct customer groups (Target/VIPs, Savers, Spenders, Budget-conscious, and Standard), enabling tailored marketing campaigns and optimized outreach.

2.1.3 DANA Indonesia (Data Warehouse + RFM + K-Means)

Corporate Profile: Financial Technology (Fintech) digital wallet operating a B2C model. *Business Pain Points:* Simple heuristic rules failed due to highly skewed transactional data, and they lacked an automated data warehouse architecture to daily ingest, standardize, score, and store customer segments. *Solution:* Built an end-to-end customer segmentation pipeline utilizing Alibaba Cloud's MaxCompute (Data Warehouse) and Machine Learning Platform for AI (PAI), applying log transformations, K-Means clustering ($k = 5$), and daily automated batch serving [**Pratama2020**]. *Results:* Enabled seamless tracking of historical behavioral shifts over time and allowed marketing teams to query user segments directly via SQL and dashboards.

2.2 Comparative Analysis Matrix

Table 2.1: Comparative Analysis Matrix

Methodology	**Nature**	**Complexity**	**Key Advantages**
RFM Segmentation	Rule-based classification (Recency, Frequency, Monetary).	Low, calculated using simple SQL/BI logic, no ML required.	Easy to implement, low cost, suitable for re-engaging direct marketing campaigns.
K-Means Clustering	Unsupervised machine learning (Unsupervised ML) using features like Annual Income and Spending Score to find natural customer groupings.	Medium, requires applying the Elbow Method to find $k = 5$ clusters and scaling feature data before processing.	Discovers hidden behavioral patterns based on income and spending habits (e.g., Target VIPs vs. High Spenders).
Data Warehouse + RFM + K-Means	Integrates RFM segmentation with K-Means clustering on a Data Warehouse platform.	High, requires skewed data transformation pipelines (log transform/standardization) and daily automated snapshot saving.	Automated segmentation for millions of users, allowing easy querying of behavioral shifts over time.

2.3 Problem Statement

Currently, CompanyX lacks a comprehensive, integrated customer analytics framework to track purchasing patterns and profile customer behaviors. Based on empirical transactional analysis, the primary strategic goal is to formulate a data-driven solution addressing three critical challenges:

- **Customer Retention & Churn Mitigation:** How can CompanyX re-engage inactive customers and effectively retain the 62.9% majority of one-time buyers who drop off after their initial order?
- **Cross-Selling Optimization:** How can we eliminate cross-selling inefficiencies and drive basket expansion for over 9,000 B2C customers who purchased bikes but never bought complementary accessories or apparel?
- **Data-Driven Segmentation Framework:** How can we establish a scalable Data Warehouse and machine learning pipeline leveraging K-Means Clustering to automatically segment CompanyX's customer base into distinct, actionable value tiers and behavioral profiles?

Chapter 3

Datawarehose Design

We propose our data warehouse design as below:

At a first glance, we can observe that the blueprint follows a Galaxy Schema (Fact Constellation) which consists of two fact tables, namely **FactCustomerSnapshot**, **FactCustomerRFMLive**; four dimensions including DimModelConfig, DimClusterProfiling, DimDate, DimCustomerProfiling; and an additional `CustomerProfile` staging table for incremental loading process (using T-SQL Merge). We discuss in details about these elements later in this chapter.

3.1 Staging Layer

The table's primary key is the combination of **BatchID** and **CustomerKey_Str**. Conceptually, this relational table acts as a Message Queue, which stores and prepares the demographics data of customers before updating or inserting necessary entries into DimClusterProfiling and DimCustomerProfiling. Moreover, with the help of batching, this table enhances the fault tolerance property of our system. For example, we can monitor the status of a specific batch with ease and rerun it in case that our ETL process encounters problems like crashes and power outage.

3.2 Dimension Layer & Slowly Changing Dimension Strategy

3.2.1 DimCustomerProfiling

We designed this dimension so that it stores customers' personal information that does not change(SCD type 0), rarely change or their historical records are not worth being put into storage (SCD type 1). These are: birth date, gender, marital status, total number of Children, Home Owner status, and total number of cars owned by that customer. And since one distinct

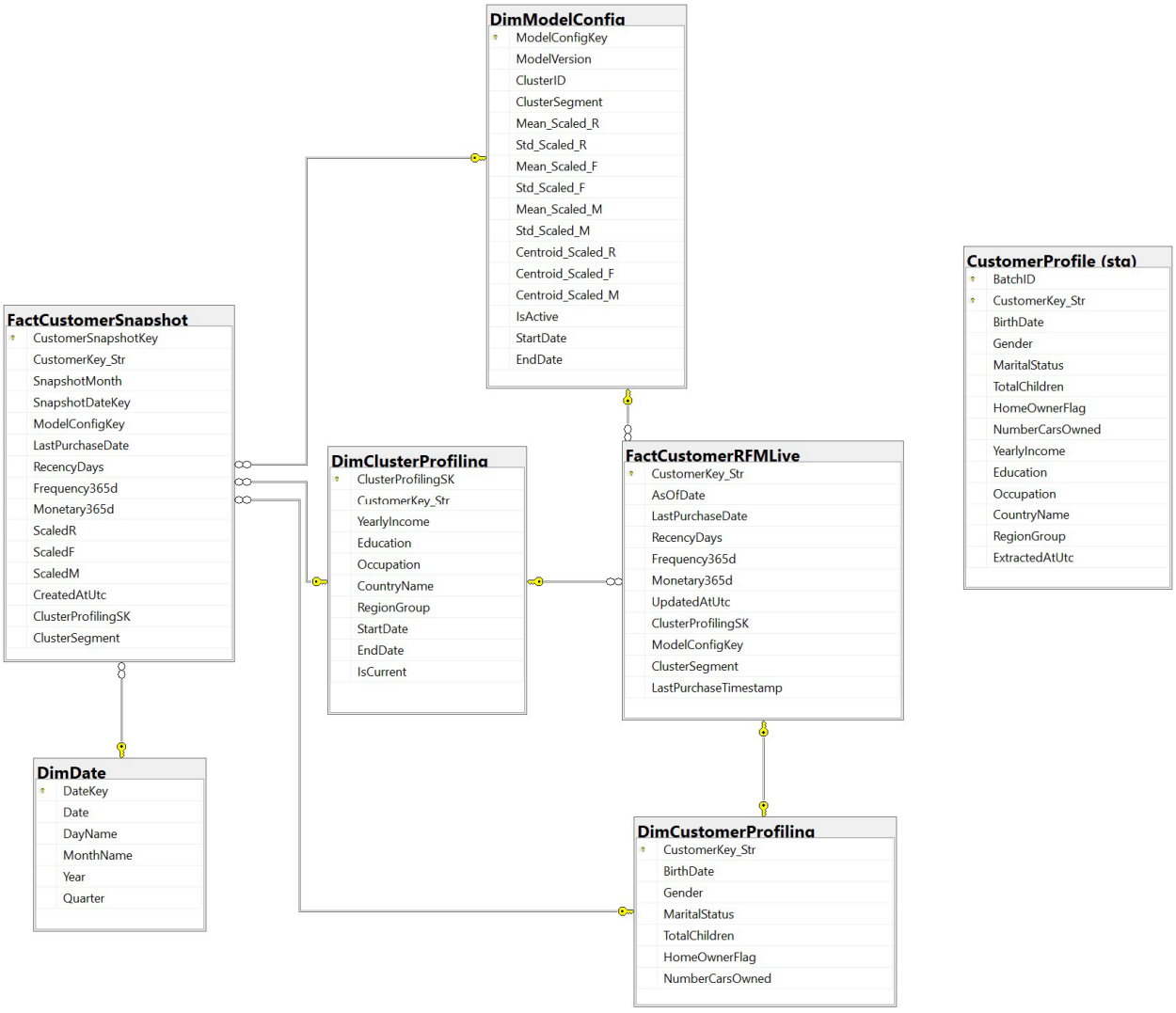


Figure 3.1: Data Warehouse Design.

customer corresponds to one row, we converted CustomerId from CompanyX.Sales.Customer to string and reuses it as the Primary key of this dimension.

3.2.2 DimClusterProfiling

Regarding this table, socioeconomic and geographical details are stored in order to complement with DimCustomerProfiling to provide analysts with a comprehensive view of customers. These details include YearlyIncome, Education, Occupation, CountryName, RegionGroup. What's more, DimClusterProfiling is designed with SCD type 2, allowing analysts to correlate whether a customer's migration to a higher-value RFM segment was preceded by change in these attributes.

3.2.3 DimModelConfig

This is created in order to track the metrics derived from our training models. These numbers will be then used to calculate the customer segment on-the-fly once one customer makes a transaction and monthly when we append our **FactCustomerSnapshot** with new rows. This will be discussed further in the next chapter. Besides that, we decided to implemenet SCD type 2 tracking for this dimension to perserve historical truth and guarantee auditability of past decisions.

3.2.4 DimDate

Each row in this table corresponds to a specific date. We planned to make this dimension static, so the date range starts from the start of the CompanyX database and ends with our current date in 2026 plus 20 years more. This ensures that if the database continues to operate, DimDate does not need to be modified in the next 20 years.

3.3 Fact Layer

3.3.1 FactCustomerSnapshot

This is a periodic snapshot fact table, which means that it will be appended with new rows in a defined schedule. In our design, FactCustomerSnapshot will be updated monthly (at the end of each month). The number of appended rows are equal to that of FactCustomerRFMLive at that specific point of time. Regarding the RFM scores(RecencyDays, Frequency365d, Monetary365d), as their names suggest, those metrics are calculated within the 365-day window.

Using this window is necessary to avoid sensonality bias and match the real world inter-purchase interval.

3.3.2 FactCustomerRFMLive

This fact table will update everytime a customer makes a transaction. However, it is different from a traditional transaction table in the sense that instead of only inserting, this table will act differently depending the customer. If he/she is the first time purchasing items, it will append the record. Otherwise, his/her old row will get overwritten by RFM values and customer segmetn derived from the new transaction. By having two fact tables as presented, we achieve separation of concerns by storing the historical records in one table and keeping up with the new real-time updates in another. This is critical when we need to compare customer segmentation over years or months and at the same time manage to take actions based on the up-to-date transactional/reporting performance.

Chapter 4

System Workflow

4.1 The General View

Before diving into the complex ETL pipeline, readers should examine the general workflow of our system as follows:

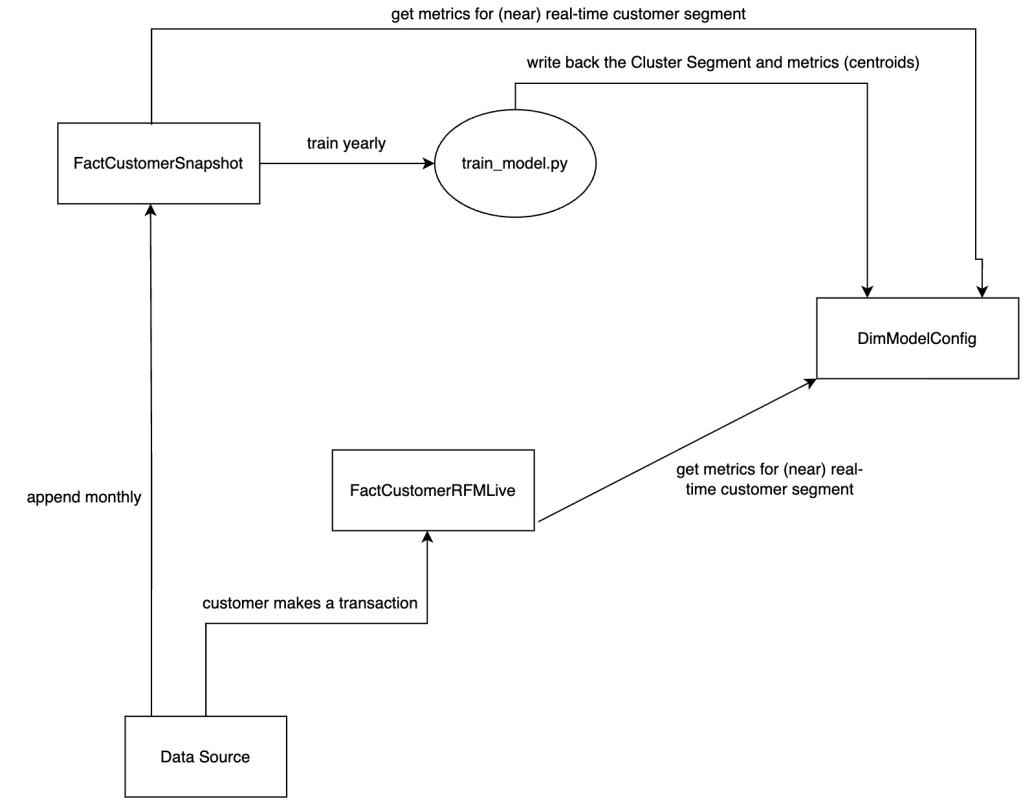


Figure 4.1: System Workflow: the image presents a high-level procedure about how transactions made by customers are processed.

At the bottom of the figure, two different paths can be observed. The first process starts with

customers purchasing items from CompanyX. Then, those transactions go through our ETL steps before reaching the data warehouse and are stored in **FactCustomerRFMLive**. Regardless of which operation(insert, update) is performed (as mentioned in 3.3.2), the results are a new R, F, and M value with updated last purchase date, and remarkably a (near) real-time clustering. This is done by using metrics stored in DimModelConfig to normalize RFM before computing it Euclidean distances against centroid values to finalized the cluster segment.

The second process happens at the end of each month where customers' RFM values at that specific date(snapshot) are captured and inserted into the **FactCustomerSnapshot**. Similar to the **FactCustomerRFMLive**, we get the cluster segment from the DimModelConfig. Whatś more, the data in this fact table will be fetched annually by our data scientists python codebase to retrain our K-means model. Following that, These new metrics are written back to the DimModelConfig and act as the active model version for the next year's customer segment.

Finally, the BI developer will take information from the data warehouse to construct meaningful charts.

4.2 ETL Pipeline & Data Flow

We present the readers with control flows of the system inside SSIS package. For further details, please refer to our submitted system resources.

4.2.1 Customer Control Flow

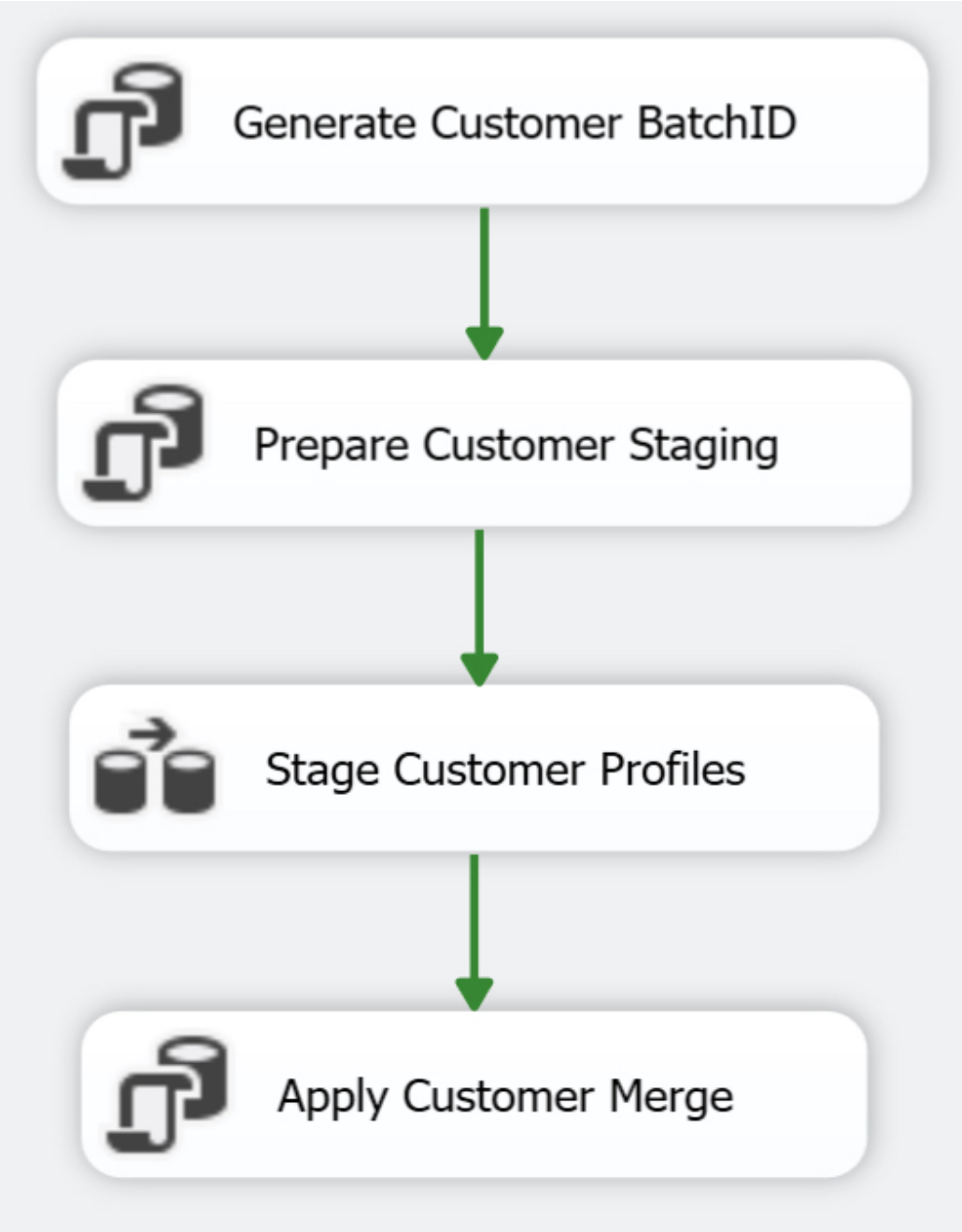


Figure 4.2: DimClusterProfiling and DimCustomerProfiling handling.

In the above figure, we show the overview of DimClusterProfiling and DimCustomerProfiling creation. First, a SQL task is executed to generate new bacht ids:

```
SELECT CONVERT(VARCHAR(36), NEWID()) AS BatchID;
```

Listing 4.1: Batch ID Generation

After that, we clean the contents in previous runs in Preparing Customer Staging:

```sql
TRUNCATE TABLE CompanyX_DataWarehouse.stg.CustomerProfile;
```

Listing 4.2: Customer Profile Truncation

Following up, we examine the data flow inside Stage Customer Profiles:

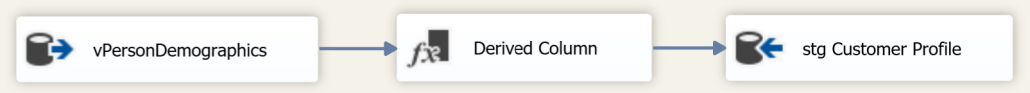


Figure 4.3: Data Flow of Stage Customer Profiles.

Overall, there are three components:

- vPersonDemographics: is the Source OLE DB block, which is used to query data from CompanyX database. After our careful data exploration, we came up with the SQL command:

```sql
SELECT
    c.CustomerID AS CustomerKey,
    d.BusinessEntityID,
    CAST(d.BirthDate AS DATE) AS BirthDate,
    d.Gender,
    d.MaritalStatus,
    d.YearlyIncome,
    d.TotalChildren,
    d.Education,
    d.Occupation,
    CAST(d.HomeOwnerFlag AS TINYINT) AS HomeOwnerFlag,
    d.NumberCarsOwned,
    CAST(d.DateFirstPurchase AS DATE) AS DateFirstPurchase,
    st.[Name] AS CountryName,
    st.[Group] AS RegionGroup
FROM Sales.vPersonDemographics AS d
INNER JOIN Sales.Customer AS c
    ON d.BusinessEntityID = c.PersonID
LEFT JOIN Sales.SalesTerritory AS st
    ON c.TerritoryID = st.TerritoryID
LEFT JOIN CompanyX_DataWarehouse.dbo.DimCustomerProfiling AS dc
    ON dc.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
LEFT JOIN CompanyX_DataWarehouse.dbo.DimClusterProfiling AS dp
    ON dp.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
   AND dp.IsCurrent = 1
```

```
26 WHERE d.DateFirstPurchase IS NOT NULL
27   AND (
28        dc.CustomerKey_Str IS NULL
29        OR dp.CustomerKey_Str IS NULL
30        OR EXISTS (
31             SELECT d.Gender, d.MaritalStatus, d.TotalChildren,
32                    CONVERT(BIT, d.HomeOwnerFlag), d.
    NumberCarsOwned
33             EXCEPT
34             SELECT dc.Gender, dc.MaritalStatus, dc.TotalChildren,
35                    dc.HomeOwnerFlag, dc.NumberCarsOwned
36        )
37        OR EXISTS (
38             SELECT d.YearlyIncome, d.Education, d.Occupation,
39                    st.[Name], st.[Group]
40             EXCEPT
41             SELECT dp.YearlyIncome, dp.Education, dp.Occupation,
42                    dp.CountryName, dp.RegionGroup
43        )
44   );
45
```

Listing 4.3: Table Selection

To summarize, the script selects customers' demographics data that needs to be be added or updated in the data warehouse. Interestingly, we have the WHERE clause which filters for customers with purchases who either do not exist in the warehouse or have outdated details. By utilizing EXCEPT clauses inside EXISTS conditions, it efficiently detects any changes in key attributes—such as personal profile data or geographical income data—acting as an incremental data load filter.

- Derived Column: cleans input data in case there are anomalies Notably, we focused on handling NULL values with the decision that it would be converted to default fallback such as "Unknown", 0, or "U"
- stg Customer Profile: creates the table 3.1 to prepare for Merge process

Finally, the process ends with the SQL task Apply Customer Merge, which is vital to apply SCD tracking to each dimension:

```
1 DECLARE @BatchID UNIQUEIDENTIFIER = CONVERT(UNIQUEIDENTIFIER, ?);
```

Derived Column Name	Derived Column	Expression	Data Type	Length
CustomerKey_Str	<add as new column>	(DT_WSTR,20)CustomerKey	Unicode string [DT_WSTR]	20
BatchID	<add as new column>	(DT_GUID)("{" + @[User::BatchID] + "}")	unique identifier [DT_GUID]	
MaritalStatus	Replace 'MaritalStatus'	ISNULL(MaritalStatus) ? "U" : (UPPER(TRIM(MaritalStatus)) == "M" || UPPER(TRIM(MaritalStatu...	Unicode string [DT_WSTR]	1
YearlyIncome	Replace 'YearlyIncome'	ISNULL(YearlyIncome) || TRIM(YearlyIncome) == "" ? "Unknown" : TRIM(YearlyIncome)	Unicode string [DT_WSTR]	30
Gender	Replace 'Gender'	ISNULL(Gender) ? "U" : (UPPER(TRIM(Gender)) == "M" || UPPER(TRIM(Gender)) == "F" ? UPP...	Unicode string [DT_WSTR]	1
CountryName	Replace 'CountryName'	ISNULL(CountryName) || TRIM(CountryName) == "" ? "Unknown" : TRIM(CountryName)	Unicode string [DT_WSTR]	50
RegionGroup	Replace 'RegionGroup'	ISNULL(RegionGroup) || TRIM(RegionGroup) == "" ? "Unknown" : TRIM(RegionGroup)	Unicode string [DT_WSTR]	50
TotalChildren	Replace 'TotalChildren'	ISNULL(TotalChildren) || TotalChildren < 0 ? 0 : TotalChildren	four-byte signed integer [DT_I4]	
Education	Replace 'Education'	ISNULL(Education) || TRIM(Education) == "" ? "Unknown" : TRIM(Education)	Unicode string [DT_WSTR]	30
Occupation	Replace 'Occupation'	ISNULL(Occupation) || TRIM(Occupation) == "" ? "Unknown" : TRIM(Occupation)	Unicode string [DT_WSTR]	30
HomeOwnerFlag	Replace 'HomeOwnerF...	ISNULL(HomeOwnerFlag) ? (DT_UI1)0 : (HomeOwnerFlag ? (DT_UI1)1 : (DT_UI1)0)	Boolean [DT_BOOL]	
NumberCarsOwned	Replace 'NumberCarsO...	ISNULL(NumberCarsOwned) || NumberCarsOwned < 0 ? 0 : NumberCarsOwned	four-byte signed integer [DT_I4]	
BirthDate	Replace 'BirthDate'	ISNULL(BirthDate) ? (DT_DBDATE)"1900-01-01" : BirthDate	database date [DT_DBDATE]	

Figure 4.4: Derived Column: input data cleaning step.

```
IF EXISTS (
    SELECT 1
    FROM stg.CustomerProfile
    WHERE BatchID = @BatchID
)
BEGIN
    EXEC dbo.usp_ApplyCustomerProfileBatch @BatchID = @BatchID;
END;
```

Listing 4.4: Merge

4.2.2 RFM Live

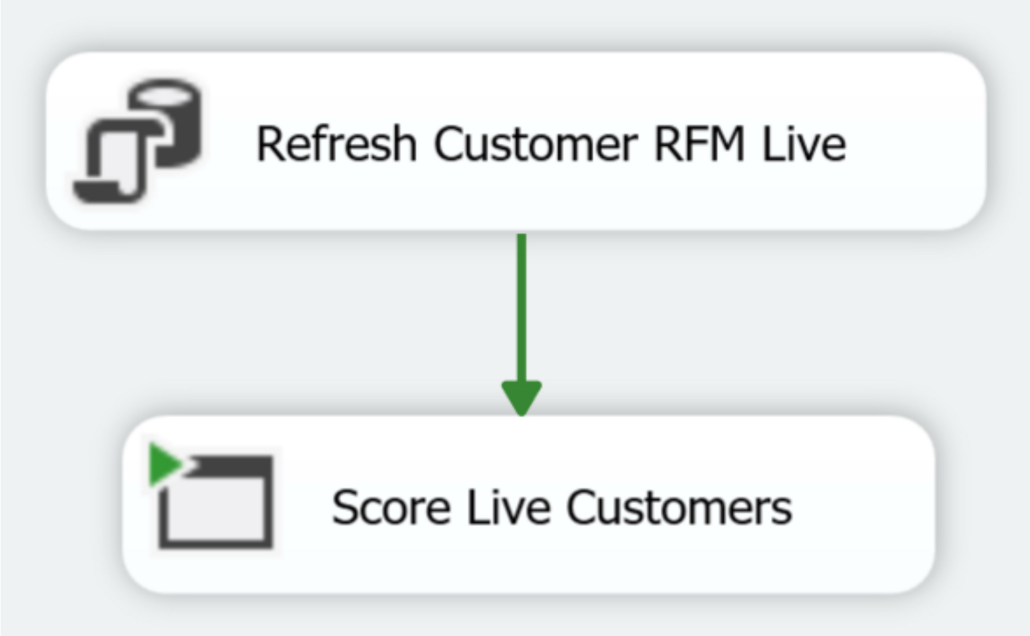


Figure 4.5: FactCustomerRFMLive handling.

pass

4.2.3 RFM Snapshot

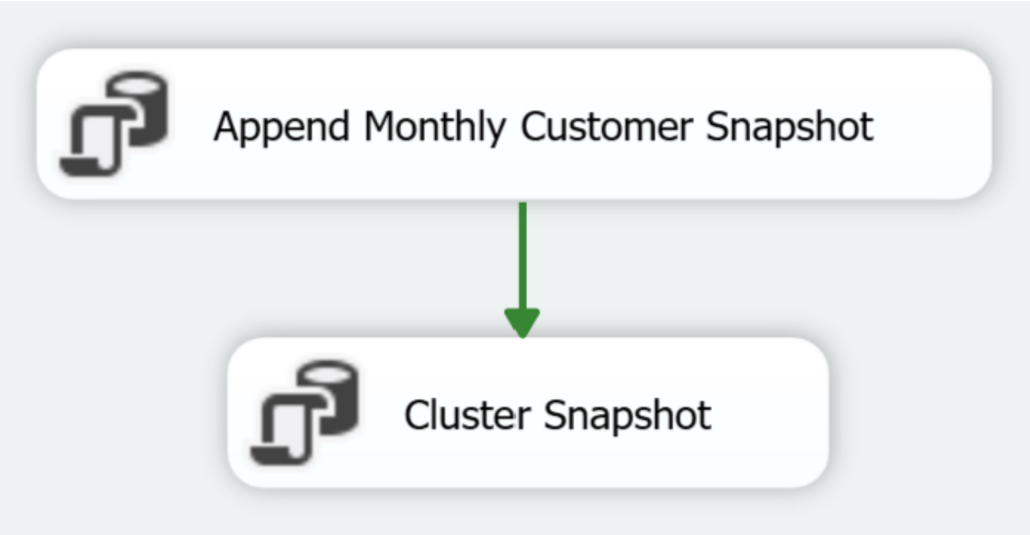


Figure 4.6: FactCustomerSnapshot handling.

The first block Append Monthly Customer Snapshot will calculate the RFM values monthly for the **FactCustomerSnapshot**:

```
DECLARE @BatchID UNIQUEIDENTIFIER = CONVERT(UNIQUEIDENTIFIER, ?);
DECLARE @SnapshotDate DATE;

SET @SnapshotDate = EOMONTH(GETDATE(), -1);

EXEC dbo.usp_AppendMonthlyRFMSnapshot
    @SnapshotDate = @SnapshotDate;
```

Listing 4.5: Monthly RFM Calculation

Next, the second block Cluster Snapshot will determine the cluster segment:

```
EXEC dbo.usp_ClusterCustomerRFMSnapshot;
```

Listing 4.6: Monthly Cluster Segment

Chapter 5

RFM Analysis and K-Means Clustering

5.1 Motivation

After investigation in the CompanyX database from 2011 to 2014, it is easy to state a problem, which is the absence of identifying and categorizing customers. Without targeted marketing strategies to customers, it is difficult to retain customers, leading to the loss in profit and revenue.

To address the problem of unstructured customer data, this project implements the RFM method, which summarizes each customer's buying behaviour with three variables, and groups customers with K-Means Clustering. This analysis enables data-driven segmentation, facilitating strategies such as targeted marketing, retention campaigns, or loyalty programmes aligned with actual customer behaviour. The data comes from the sales records of CompanyX, covering from May 2011 to December 2014.

5.1.1 RFM variables

Recency (R): the number of days since the customer's last purchase.

Frequency (F): the total number of purchases the customer has made.

Monetary (M): the total amount the customer has spent up to the snapshot date.

5.2 Data Cleaning

The raw data is first ingested from the FactCustomerSnapshot, and then processed through deduplication. After that, any R, F, or M values are missing, the process stops and reports the number of affected rows to the ETL Team. Anomalies values such as negative values are set to zero, since a purchase count or a monetary value can not be negative.

This subset of data serves a representative overview of customer purchases based on RFM analysis as mentioned above.

Customer	Recency	Frequency	Monetary
11057	37	1	3729.36
11058	22	1	3756.99
11060	12	1	3756.99
11061	18	1	3729.36
11072	0	1	3756.99
11238	84	1	3756.99
11239	71	1	3729.36
11240	32	1	3756.99
11243	38	1	3756.99
11244	47	1	3756.99
11247	41	1	3729.36
11250	14	1	3756.99
11333	10	1	3756.99
11334	27	1	3756.99
11337	22	1	3729.36
11591	79	1	3953.99
11592	79	1	3953.99
11593	69	1	3953.99
11599	72	1	3953.99
11601	71	1	3953.99
11606	82	1	3953.99
11607	71	1	3953.99

Table 5.1: RFM analysis

5.3 Log transformation

Spending and purchase counts are usually right-skewed. Most customers spend a moderate amount, while only a small number spend many times more. If raw values are used, these large values stretch the scale and the K-Means model spends more effort unnecessarily separating a few high spenders (outliers) rather than grouping typical customers. By using log transformation, the issue can be minimized.

For example, a customer who spends 3,000 and 300,000 differs by a factor of 100 in raw terms, but their log values differ by only 4.6. We use log(1+x) in this case, written log1p

because some customers have a value of zero, which leads to log(0) undefined.

Customer	Recency	Frequency	Monetary
11057	3.638	0.693	8.224
11058	3.135	0.693	8.232
11060	2.565	0.693	8.232
11061	2.944	0.693	8.224
11072	0.000	0.693	8.232
11238	4.443	0.693	8.232
11239	4.277	0.693	8.224
11240	3.497	0.693	8.232
11243	3.664	0.693	8.232
11244	3.871	0.693	8.232
11247	3.738	0.693	8.224
11250	2.708	0.693	8.232
11333	2.398	0.693	8.232
11334	3.332	0.693	8.232
11337	3.135	0.693	8.224
11591	4.382	0.693	8.283
11592	4.382	0.693	8.283
11593	4.248	0.693	8.283
11599	4.290	0.693	8.283
11601	4.277	0.693	8.283
11606	4.419	0.693	8.283
11607	4.277	0.693	8.283

Table 5.2: Logged Data

5.4 Standardization

Before importing data to run the K-Means Clustering model, it is essential to standardize the RFM values to ensure that each metric contributes equally to the analysis, since Recency, Frequency, and Monetary values are measured on different scales. K-Means relies heavily on distances (later will be shown), so a variable with a larger spread would dominate the result.

StandardScaler solves this by subtracting the mean and dividing by the standard deviation of each variable, so that each one has a mean of 0 and a standard deviation of 1. The mean and standard deviation are estimated from the training data and stored, so that new customers are scaled in the same way.

Customer	Recency	Frequency	Monetary
11057	0.109	0.000	-0.875
11058	-0.392	0.000	-0.581
11060	-0.962	0.000	-0.581
11061	-0.583	0.000	-0.875
11072	-3.522	0.000	-0.581
11238	0.913	0.000	-0.581
11239	0.747	0.000	-0.875
11240	-0.032	0.000	-0.581
11243	0.135	0.000	-0.581
11244	0.342	0.000	-0.581
11247	0.209	0.000	-0.875
11250	-0.819	0.000	-0.581
11333	-1.128	0.000	-0.581
11334	-0.196	0.000	-0.581
11337	-0.392	0.000	-0.875
11591	0.852	0.000	1.454
11592	0.852	0.000	1.454
11593	0.719	0.000	1.454
11599	0.761	0.000	1.454
11601	0.747	0.000	1.454
11606	0.889	0.000	1.454
11607	0.747	0.000	1.454

Table 5.3: Standardized Data

5.5 K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to partition data into a dataset into K distinct, non-overlapping clusters. The purpose of this part is to find the most optimal K, meaning that the optimal number of groups categorizing customers.

For each data point in the dataset, calculate the Euclidean distance between the point and each centroid. Assign the data point to the cluster with the closest centroid

$$d(x, y) = \sqrt{\sum_{n}(y_i - x_i)^2}$$

Where:

- $d(x, y)$: the distance between data point x and centroid y
- y_i: donates the value of the i-th attribute value of centroid y
- x_i: represents the i-th attribute of the data point x
- i : refers to index of the attribute
- n : indicates the total number of the attribute for each data point

The WCSS is calculated to measure the compactness of the clusters. For each cluster, compute the sum of squared distances between each data point and its cluster centroid(Kuraria et al., 2018),(Cui, 2020):

$$WCSS = \sum\sum\sum(x_{ij} - \bar{x}_{kj})^2$$

Where:

- K: Total number of clusters.
- k: An index that refers to the current cluster, ranging from 1 to K.
- S_k: The set of data points assigned to cluster k.
- p: Number of features or dimensions in the dataset.
- j: An index that refers to the current feature or dimension, ranging from 1 to p.
- x_{ij}: The j-th feature value of the i-th data point.
- $\bar{x}_{kj}$: The mean value of the j-th feature in cluster k (i.e., the centroid's value).

- $(x_{ij} - \bar{x}_{kj})^2$: The squared Euclidean distance between the j-th feature of data point xi and the corresponding feature of the centroid of its cluster.

This WCSS can also be known as the Inertia, the lower means the tighter the data points are grouped. Another important criteria, which is Silhouette, also needs to be considered. The silhouette score measures how well-separated and cohesive clusters are in K-Means clustering. It evaluates each data point's fit relative to its own cluster versus neighboring clusters.

Graph: Elbow Method to find optimal K

Graph: Silhouette Score based on different K

Prior to clustering, the optimal number of clusters (k) is determined by the Elbow Method, which plots the Within-Cluster Sum of Squares (WCSS) against different values of k. The point at which the rate of decrease sharply (the elbow) is identified at k = 5, suggesting that five clusters offer a good balance.

Then the standardized RFM data is then fit to the standardized RFM Model, resulting in cluster labels among customers. The results are visualized using 2D scatter plots of RFM components, colored by cluster.

Graph: Customer Clusters

K Means only returns a number of each customer (0 to 4), which reflects no value of what the groups mean. Moreover, annually the model is retrained and the number from 0 to 4 is different and not staying consistent. Therefore, to turn them into meaningful clusters automatically, a solution is implemented: Each cluster on an RFM variable is given "points", based on how its average compares with the other clusters. The rules are

For Monetary and Frequency: the cluster with the highest average gets 5 points and the lowest get 1. (the higher these values are, the more valuable they are)

For Recency: the rule is reversed because shorter time since the last purchase is better.
Formula: Score = Monetary Points + Frequency Points + Recency Points x 1.1

Recency is weighted slightly higher, because a recent purchase is a strong sign that a customer is still active. Points are given to clusters, not to individual customers.

The outcome will look like this:

The total is 18,484 customers, which matches the number of rows in the last date of the database.

Champions (cluster 4) has the highest frequency and spend, with an average of 2.55 purchases and 1,694 in total. Its average recency is 110 days, so these customers are valuable but have not bought recently.

Cluster	Avg Recency	Avg Frequency	Avg Monetary	R points	F points	M points	Score	Segment
4	110.34	2.55	1,694.00	4	5	5	14.4	Champions
2	13.31	1.24	96.10	5	4	3	12.5	Recent / Promising
0	174.17	1.00	1,518.38	3	2	4	9.3	High-Value / Cooling Off
3	196.03	1.03	48.91	2	3	2	7.2	Low-Spend / Occasional
1	602.92	0.00	0.00	1	1	1	3.1	Lost / Dormant

High-Value / Cooling Off (cluster 0) is the second-largest group by spend, but each customer has bought only once on average. Their last purchase was about 174 days ago, so they are at risk of leaving.

Low-Spend / Occasional (cluster 3) is the largest group. Customers buy about once, spend about 49 on average, and last bought about 196 days ago.

Recent / Promising (cluster 2) bought most recently, with an average of 13 days since the last purchase. Their spend is low so far, which suggests they are new customers who may grow.

Lost / Dormant (cluster 1) has no recorded frequency or spend in the period, and an average recency of 603 days. These customers have stopped buying.

5.6 Conclusion

The result of the RFM Analysis and K-Means clustering reveal the distinct customer segments with unique purchasing behaviours. By standardizing the RFM variables and applying the K-Means algorithm, the dataset was successfully divided into five interpretable clusters. Such segmentation highlights not only the diversity in customer engagement levels but also provides valuable insights for strategic business decision-making. Therefore, the CompanyX will be able to design and launch effective marketing strategies to each cluster of customers, not only to retain the occasional buyers, but also to come up with attractive promotions for recent customers.

Reference:

Kuraria, A., Jharbade, N., & Soni, M. (2018). Centroid Selection Process Using WCSS and Elbow Method for KMean Clustering Algorithm in Data Mining. International Journal of Scientific Research in Science, Engineering and Technology, 6(1), 190–195. https://doi.org/10.32628/ijsrse

Chapter 6

Business Intelligence & Analytics Dashboards

6.1 Context & Dashboard Objectives

6.1.1 Business Context

After completing the data warehouse and running the K-Means clustering model, our team needed an interactive tool to show the results to decision-makers. In the database, customer data and cluster numbers are stored in tables, which are difficult for managers to read and understand quickly. Therefore, we built a Business Intelligence (BI) dashboard using Microsoft Power BI.

The dashboard connects directly to the dimensional data warehouse and focuses on three main goals:

1. **Executive Overview:** Help company directors monitor total sales, customer numbers, yearly trends, and early warnings about customer churn.

2. **Customer Demographics:** Help marketing and sales teams understand customer profiles, such as location, income level, age group, gender, and job types.

3. **RFM Segmentation and Strategy:** Help the CRM team see how each customer cluster behaves in terms of Recency, Frequency, and Monetary value, and choose the right marketing strategy for each group.

6.1.2 Visualization Principles from the Course

To make sure the dashboard is easy to read and follows good data visualization rules, we applied the guidelines from Wilke [1] and Dr. Phan Trong Nhan's lecture slides:

- **Treemaps for Categories (Slide 31):** As taught in class, pie charts with more than 3 or 4 slices are difficult to read. We used treemaps to show customer occupation and segment breakdown clearly.
- **100% Stacked Column Charts for Proportions over Time (Slide 50–51):** We used this chart type to compare how the percentage of each customer segment changed from 2011 to 2014.
- **Histograms for Distributions (Slide 54–55):** We grouped customer ages into clean bins (< 30, $30 - 40$, $41 - 50$, $51 - 60$, $60+$) to show the age distribution of our buyers.
- **Bubble Chart for Multi-Variable Associations (Slide 57–59):** We plotted Recency on the X-axis, Monetary on the Y-axis, customer count as bubble size, and cluster colors to show the relationship between multiple metrics.
- **Consistent Colors across Pages (Slide 21):** We kept the exact same 5 colors for the 5 customer segments across all 3 pages so viewers do not get confused:
  - **1. Champions:** Red (#A82020)
  - **2. High-Value / Cooling Off:** Dark Green (#22593D)
  - **3. Recent / Promising:** Orange (#D97F1A)
  - **4. Low-Spend / Occasional:** Teal (#259696)
  - **5. Lost / Dormant:** Purple (#4C2773)

6.1.3 Dashboard Structure

The report is organized into 3 main pages:

1. **Page 1: Executive Customer & Segmentation Overview** (High-level business health and trends).
2. **Page 2: Customer Demographics & Persona** (Detailed customer background information).
3. **Page 3: Customer RFM Segmentation & Behavior** (Detailed cluster metrics and strategic matrix).

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6.2 Page 1: Executive Customer & Segmentation Overview

Page 1 is designed for high-level managers who need a quick summary of the company's customer base, revenue, and retention status.

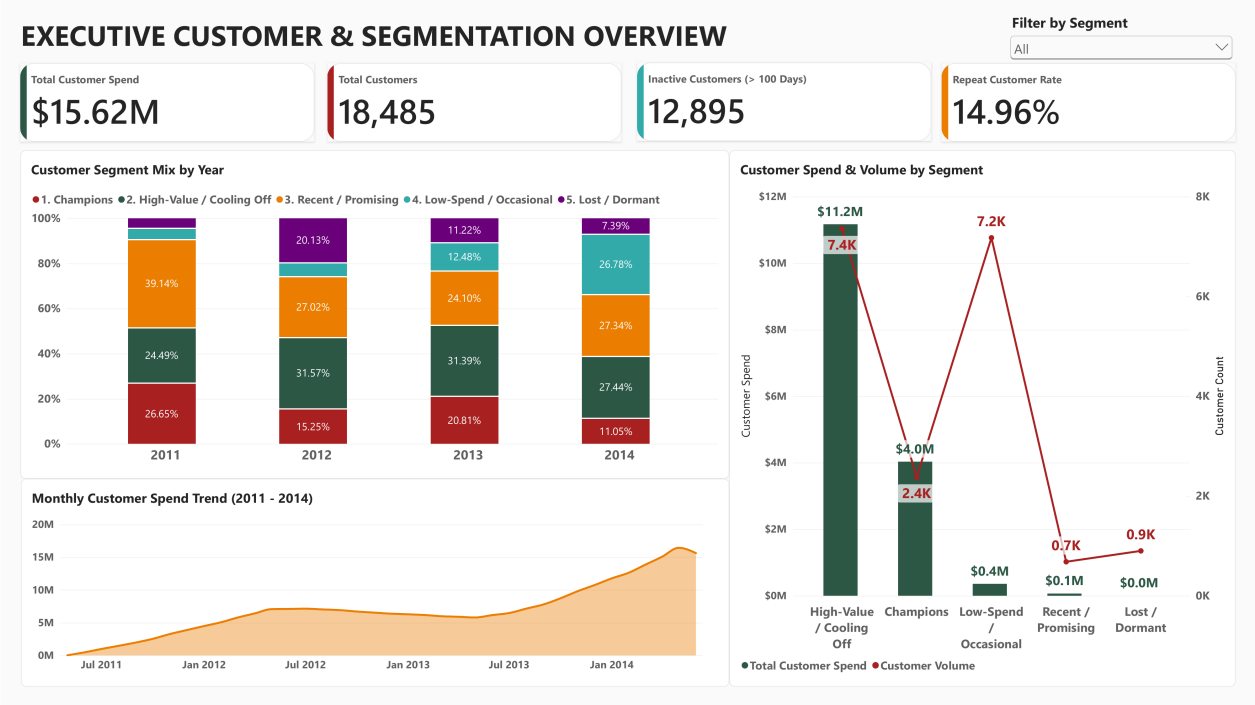


Figure 6.1: Executive Customer & Segmentation Overview Dashboard (Page 1).

6.2.1 Core Executive KPIs

At the top of Page 1, we place four key metrics:

1. **Total Customer Spend ($15.62M):** Total lifetime sales from all individual retail customers ($15,615,354.05).

2. **Total Customers (18,485):** Total number of unique customers in our database.

3. **Inactive Customers (> 100 Days) (12,895):** The number of customers who have not made any purchases in the last 100 days. This makes up **69.76%** of all customers. In the bicycle retail industry, bikes are durable goods, so people do not purchase every month. However, after 100 days (more than 3 months), if a buyer does not come back for small accessories or maintenance, they are at risk of leaving. Therefore, this KPI serves as an early-warning signal.

4. **Repeat Customer Rate (14.96%):** Calculated as the percentage of customers who have ordered at least two times:

$$\text{Repeat Customer Rate} = \frac{\sum[\text{Customers with Frequency} \geq 2]}{\text{Total Customers}} = \frac{2{,}765}{18{,}485} = 14.96\% \tag{6.1}$$

This shows a serious business issue: **85.04% of customers only bought once and never returned**.

6.2.2 Visual Charts and Key Findings

1. Customer Segment Mix by Year (2011–2014)

We selected a 100% Stacked Column Chart to show the proportion of each customer segment over the four operational years:

- **2011 Baseline:** In the first year, CompanyX focused on establishing its retail presence. *Recent / Promising* customers constituted the largest group at **39.14%**, followed by *Champions* at **26.65%** and *High-Value / Cooling Off* at **24.49%**. The *Low-Spend / Occasional* and *Lost / Dormant* segments formed only a small portion (roughly 9.7% combined).
- **2012 Transition:** As earlier buyers matured, the proportion of *High-Value / Cooling Off* customers increased to **31.57%**. However, buyers who did not place repeat orders transitioned into *Lost / Dormant*, which rose to **20.13%**. The share of *Champions* stood at **15.25%**, while *Recent / Promising* made up **27.02%**.
- **2013 Product Diversification:** In 2013, the business began offering bike accessories and apparel. As a result, the *Low-Spend / Occasional* segment grew substantially to **12.48%**. The *High-Value / Cooling Off* group remained steady at **31.39%**, and *Champions* accounted for **20.81%**.
- **2014 Mass Market Expansion:** By 2014, the proportion of *Low-Spend / Occasional* shoppers more than doubled to **26.78%**. Meanwhile, *High-Value / Cooling Off* represented **27.44%**, *Recent / Promising* was **27.34%**, and *Champions* made up **11.05%**. The *Lost / Dormant* group dropped to **7.39%**.

These figures illustrate an important strategic shift: while CompanyX was originally a niche retailer of premium bicycles dominated by high-value buyers in 2011–2012, it successfully expanded to general everyday shoppers in 2013–2014. Nevertheless, this expansion brought a large volume of one-time or low-spending buyers.

2. Monthly Customer Spend Trend (2011–2014)

We used an Area Chart on a continuous date timeline from July 2011 to June 2014 to track overall revenue progression:

- **Early Stage (2011–Mid 2012):** Monthly sales started at a moderate level, which was typical for a new business with a limited product catalog.
- **Rapid Acceleration (Mid 2012–End 2013):** Starting in the second half of 2012, sales increased significantly. The chart shows strong upward momentum throughout 2013, reaching the highest sales volume towards the end of 2013.
- **Seasonality Trends:** The area chart displays seasonal fluctuations, with higher sales during late spring and summer (peak outdoor cycling season) as well as year-end holiday shopping periods.

3. Customer Spend & Volume by Segment

This Dual-Axis Combo Chart displays total sales as bars on the left axis and customer counts as points on the right axis:

- **Validation of the Pareto Principle (80/20 Rule):**
  - The *High-Value / Cooling Off* cluster generates **$11.2M** (71.53% of total revenue) from 7,357 customers (39.80% of the customer base).
  - The *Champions* cluster contributes **$4.0M** (25.81% of total revenue) from 2,377 customers (12.86%).
  - Together, these two premium segments contribute **97.34%** of CompanyX's total lifetime sales ($15.2M out of $15.62M), while making up only **52.66%** of all customers (9,734 people).
- **The Low-Monetary Contrast:** In comparison, the *Low-Spend / Occasional* group contains 7,175 customers (38.82% of the total base) but brings in only **$0.35M** (2.25% of revenue). Similarly, *Recent / Promising* accounts for **$0.07M** (0.42%), and *Lost / Dormant* accounts for **$0.00**.
- **Key Takeaway:** This clearly demonstrates that retaining high-value customers is far more essential for business survival than simply acquiring many low-spend shoppers.

6.2.3 Dashboard Interactivity and Slicer Architecture

In our Power BI solution, an interactive dropdown slicer (*Filter by Segment*) is placed in the top-right corner of each dashboard page. However, during the dashboard development, an important architectural decision was required for Page 1:

- **The Macro Trend Conflict:** The two longitudinal visuals—*Customer Segment Mix by Year* and *Monthly Customer Spend Trend*—rely on historical snapshot data from the periodic snapshot table (`FactCustomerSnapshot`) to show the whole company's evolution. In particular, the 100% stacked column chart displays the structural proportions among all five segments. If a single segment is filtered, the multi-segment composition collapses to 100%, eliminating its analytical purpose.
- **Implementation of Edit Interactions:** Following standard Business Intelligence best practices, we accessed the *Format* → *Edit Interactions* menu in Power BI and set the interaction from the segment slicer to **None** for these two macro trend visuals. As a result, users can filter the top KPI cards and the segment combo chart without breaking the overall company baseline timeline.

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6.3 Page 2: Customer Demographics & Persona Profiling

Page 2 helps us understand who our customers are, where they live, and what their personal characteristics look like.

6.3.1 Demographic Overview KPIs

The four cards on the left show customer averages:

- **Total Customers:** 18,485.
- **Total Customer Spend:** $15.62M.
- **Average Spend per Customer: $844.76**.
- **Average Order Value (AOV): $720.53**.

Because Average Spend ($844.76) is very close to Average Order Value ($720.53), it confirms again that most people only place one single order in their lifetime.

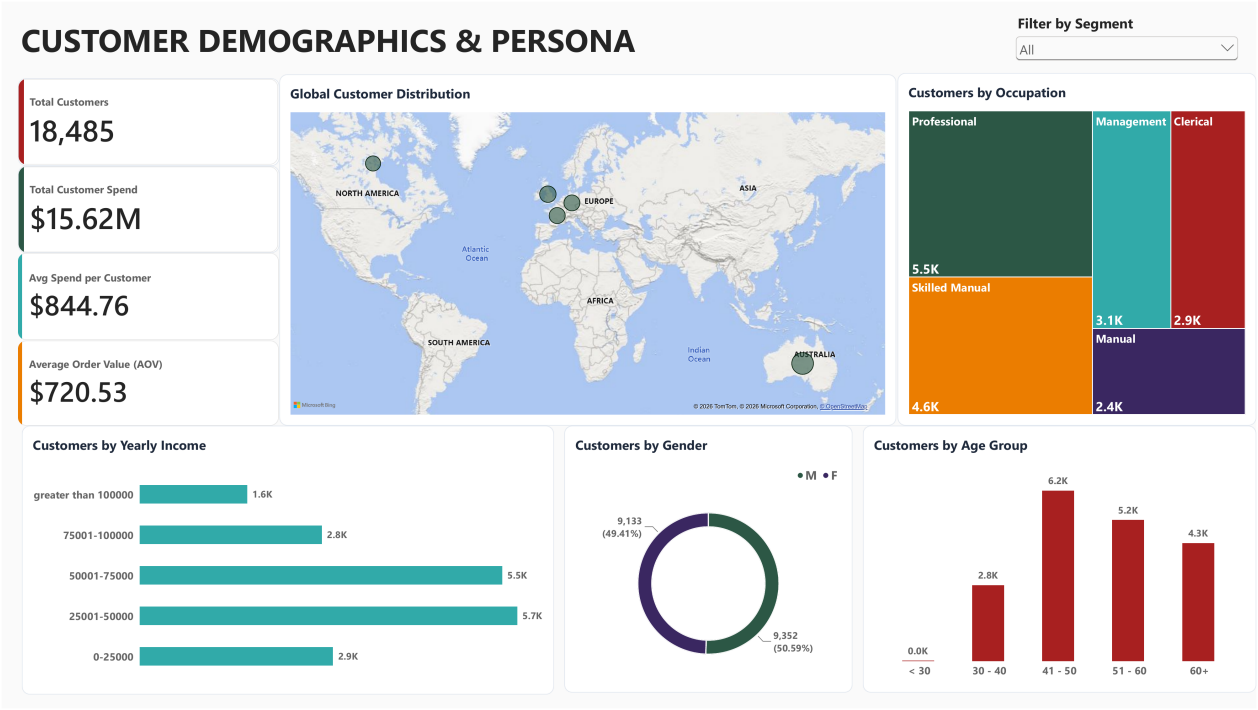


Figure 6.2: Customer Demographics & Persona Dashboard (Page 2).

6.3.2 Demographic Analysis (Overall Base)

1. Geographic Distribution (Map Visual)

The map shows that our customers live in developed markets with strong cycling cultures, mainly in North America (United States, Canada), Western Europe (United Kingdom, Germany, France), and Australia.

2. Customers by Occupation (Treemap)

- **Professional** (5,505 people, 29.8%) and **Skilled Manual** (4,577 people, 24.8%) make up more than **54.6%** of all customers.
- Other occupations include Management (3,075, 16.6%), Clerical (2,928, 15.8%), and Manual workers (2,400, 13.0%).
- This shows that our core buyers are educated workers and skilled technicians who have good income and an active lifestyle.

3. Customers by Yearly Income (Horizontal Bar Chart)

- The middle-income groups are the largest: **$25,001–$50,000** has 5,720 customers (30.9%), and **$50,001–$75,000** has 5,496 customers (29.7%).

- Together, middle-class customers make up **60.6%** of the total customer base. High earners ($> \$100,000$) represent 1,600 customers (8.7%), while entry-level earners ($0–$25,000) account for 2,870 customers (15.5%).

4. Customers by Gender (Donut Chart)

- **Male:** 9,352 customers (50.59%).
- **Female:** 9,133 customers (49.41%).
- The gender ratio is almost 50/50, which proves that cycling products are popular with both men and women equally.

5. Customers by Age Group (Column Histogram)

- Customers are grouped into age brackets: < 30, $30 - 40$, $41 - 50$, $51 - 60$, and 60+.
- The majority of customers are in middle age: **41–50 years old** (6,200 people, 33.5%) and **51–60 years old** (5,200 people, 28.1%).
- People under 30 make up a very small portion, meaning that young adults rarely buy expensive bicycles from CompanyX.

6.3.3 Segment-Specific Demographic Personas (Slicer Drill-Down)

By utilizing the interactive segment slicer on Page 2, decision-makers can drill down into the unique demographic persona of each customer cluster. Table 6.1 provides an empirical comparison across the five groups.

Table 6.1: Comparative Demographic Profiles Across Customer Segments.

Segment	Customers	Total Spend	AOV ($)	Top Occupation	Peak Income	Dominant Age
1. Champions	2,377	$4.03M	$665.99	Professional (39.2%)	$50k–$75k (30.9%)	41–50 (38.3%)
2. High-Value	7,357	$11.17M	$1,518.20	Professional (29.9%)	$50k–$75k (32.2%)	41–50 (39.4%)
3. Recent	679	$65.18K	$77.22	Skilled Manual (28.7%)	$25k–$50k (29.8%)	60+ (29.5%)
4. Low-Spend	7,175	$350.92K	$47.33	Professional (27.9%)	$25k–$50k (30.6%)	60+ (29.3%)
5. Lost / Dormant	897	$0.00	$0.00	Skilled Manual (37.8%)	$25k–$50k (43.5%)	41–50 (39.6%)

Key Findings from Segment Drill-Down

Filtering each segment on Page 2 highlights significant differences in purchasing power, age distribution, and professional background:

1. **1. Champions (Affluent White-Collar Repeat Buyers):**

- *Highest Earning Power:* Champions have the highest income level among all clusters. Buyers earning more than $100,000 account for **14.01%** (333 individuals), while 30.88% earn $50,000–$75,000.
- *White-Collar Roles: Professional* (39.17%, 931 people) and *Management* (18.93%, 450 people) represent **58.10%** of this group, demonstrating that corporate earners drive repeat purchases.
- *Distinct Repeat Behavior:* This is the only segment where Average Spend per Customer ($1,700) is much higher than Average Order Value ($665.99). This confirms that Champions make multiple repeat transactions ($F \approx 2.55$), buying replacement parts and upgrades.
- *Age Concentration:* Heavily centered in the **41–50** age bracket (**38.33%**, 911 buyers), followed by 51–60 (27.93%). These are established professionals investing in active health and cycling hobbies.

2. **2. High-Value / Cooling Off (Single-Purchase Premium Buyers):**

- *Spend Equals AOV:* Average Spend per Customer exactly equals Average Order Value at **$1,518.20**. This proves that these 7,357 customers purchased one high-end bicycle and have not yet returned.
- *Solid Middle Class:* The largest income groups are $50,001–$75,000 (32.17%) and $25,001–$50,000 (31.58%), representing **63.75%** of the cluster.
- *Age Group:* The dominant bracket is 41–50 years old (39.42%, 2.9K buyers), followed by 51–60 (28.55%, 2.1K buyers).

3. **3. Recent / Promising (Older Entrants & Skilled Technicians):**

- *Unexpected Age Shift:* In contrast to Champions and High-Value clusters, the single largest age group in *Recent / Promising* is **60+ years old** (**29.46%**, 200 customers). This demonstrates that senior riders and retirees represent a growing acquisition channel.
- *Occupation Distribution: Skilled Manual* (28.72%, 195 people) surpasses *Professional* (27.54%, 187 people) as the leading occupation.
- *Early Repeat Signals:* With an AOV of $77.22 and an average spend of $95.99, a subset of customers have already completed second accessory orders within two weeks ($R = 13.28$ days).

4. **4. Low-Spend / Occasional (Budget-Conscious Accessory Shoppers):**

- *Lower-Income Skew:* This group contains the highest percentage of lower-income shoppers: buyers earning below $25,000 account for **20.60%** (1,478 individuals).
- *Job Breakdown: Manual* workers represent 16.7% (1.2K people, tied with Management), which is the highest manual laborer proportion among all active clusters.
- *Age Profile:* Like the Recent segment, this cluster has a heavy concentration of seniors aged **60+** (**29.3%**, 2.1K buyers), alongside a noticeable group of younger adults aged 30–40 (18.1%).
- *Basket Size:* The AOV is very small at **$47.33**, confirming that these customers only buy minor accessories such as inner tubes, water bottles, and helmets.

5. **5. Lost / Dormant (Disengaged Lower-Income Manual Workers):**

- *Occupation Disparity: Skilled Manual* workers heavily dominate at **37.79%** (339 buyers), while *Management* drops to its lowest proportion at **8.70%**.
- *Compressed Earnings:* Over **80.49%** of customers earn between $25,000 and $75,000, with only 4.24% earning above $100,000.
- *Complete Inactivity:* With $0.00 in spend and over 600 days without an order, these contacts have churned from the brand.

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6.4 Page 3: Customer RFM Segmentation & Behavioral Matrix

Page 3 explains the 5 customer clusters from our K-Means model using the RFM framework (Recency, Frequency, Monetary).

6.4.1 Company-Wide Benchmark Metrics

The top ribbon gives the overall company averages:

- **Average Recency: 189.32 Days** (around 6.3 months since the last order).
- **Average Frequency: 1.17 Orders** per customer.
- **Average Monetary: $844.76** total spend per customer.

6.4.2 Cluster Performance Table

Table 6.2 shows the exact statistics for each of the 5 segments:

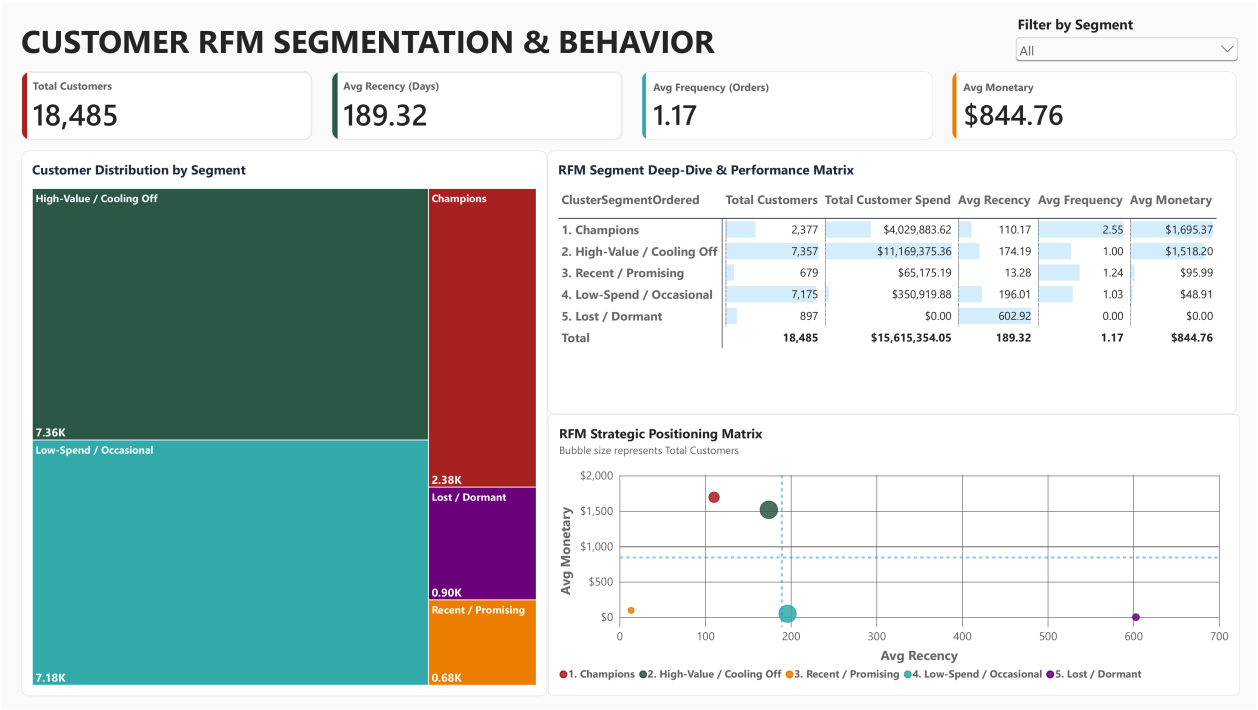


Figure 6.3: Customer RFM Segmentation & Behavior Dashboard (Page 3).

Table 6.2: Customer RFM Cluster Statistics.

Cluster Segment	**Customers**	**Total Spend ($)**	**Avg R (Days)**	**Avg F**	**Avg M ($)**
1. Champions	2,377 (12.86%)	$4,029,883.62	110.17	2.55	$1,695.37
2. High-Value / Cooling Off	7,357 (39.80%)	$11,169,375.36	174.19	1.00	$1,518.20
3. Recent / Promising	679 (3.67%)	$65,175.19	13.28	1.24	$95.99
4. Low-Spend / Occasional	7,175 (38.82%)	$350,919.88	196.01	1.03	$48.91
5. Lost / Dormant	897 (4.85%)	$0.00	602.92	0.00	$0.00
Company Total / Average	**18,485**	**$15,615,354.05**	**189.32**	**1.17**	**$844.76**

6.4.3 Customer Volume Distribution (Donut Chart)

The donut chart on the left side of Page 3 illustrates the relative share of each customer group:

- **High-Value / Cooling Off:** 7,357 customers (**39.80%** of total volume).
- **Low-Spend / Occasional:** 7,175 customers (**38.82%**).
- **Champions:** 2,377 customers (**12.86%**).
- **Lost / Dormant:** 897 customers (**4.85%**).
- **Recent / Promising:** 679 customers (**3.67%**).

This chart emphasizes that more than 78% of the customer base is divided almost evenly between two contrasting groups: high-spending buyers who need urgent retention, and low-spending buyers who make infrequent accessory purchases.

6.4.4 Strategic Positioning Matrix (Bubble Chart)

The bubble chart at the bottom-right plots all 5 clusters:

- **X-Axis:** Average Recency (from 0 to 700 days). Lower is better.
- **Y-Axis:** Average Monetary (from \$0 to \$2,000). Higher is better.
- **Bubble Size:** Total number of customers in the cluster.
- **Benchmark Lines:** Dotted reference lines at company averages ($X = 189.32$ days and $Y = \$844.76$) divide the chart into four quadrants.

Quadrant Analysis

1. **Top-Left (High Spend, Low Recency) — Champions:** These customers bought recently ($R = 110.17$ days) and spend the most ($M = \$1,695.37$). They are our best VIP customers.
2. **Top-Center (High Spend, Moderate Recency) — High-Value / Cooling Off:** These customers spent a lot ($M = \$1,518.20$), but their Recency is 174.19 days, which is close to the 189-day average. They are starting to cool down and need immediate attention before they leave.
3. **Bottom-Left (Low Spend, Very Recent) — Recent / Promising:** These are new customers who bought recently ($R = 13.28$ days). Their spend is still small (\$95.99), but they have great potential to grow into Champions.

4. **Bottom-Right (Low Spend, High Recency) — Low-Spend & Lost:** These customers have not bought for a long time and spend very little. The Lost group has not returned for over 600 days.

—

6.5 Actionable Business Recommendations

Based on the dashboard insights, we recommend the following practical strategies for CompanyX:

6.5.1 1. Champions: VIP Loyalty Program

- **Goal:** Keep our best customers happy and encourage them to recommend the brand to others.
- **Action:** Create a VIP club that gives early access to new bicycle models, invitations to special cycling events, and referral discounts when they introduce friends.

6.5.2 2. High-Value / Cooling Off: Urgent Retention Campaigns

- **Goal:** Protect the most important revenue source (over 71% of total sales).
- **Action:** Set up an automated email at day 120 after their purchase offering a free annual bicycle check-up at local partner shops, or a $150 discount voucher if they upgrade components within 30 days.

6.5.3 3. Recent / Promising: Cross-Selling and Welcome Series

- **Goal:** Encourage new buyers to make a second purchase quickly.
- **Action:** Send a 30-day email series: Day 3 with bike assembly and safety tips, Day 14 with cleaning advice, and Day 21 offering a 15% discount on essential accessories like helmets, gloves, and lights.

6.5.4 4. Low-Spend / Occasional: Low-Cost Automated Marketing

- **Goal:** Increase purchase frequency without spending too much marketing budget.

- **Action:** Avoid expensive phone calls or printed mail. Instead, send automated emails and seasonal clearance announcements for cheap consumables like water bottles and spare tubes.

6.5.5 5. Lost / Dormant: Cost Reduction

- **Goal:** Stop wasting advertising money on inactive contacts ($R > 600$ days).
- **Action:** Remove this group from paid Facebook and Google ads. Only send them one or two broad holiday sale emails per year (e.g., Black Friday or New Year).

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6.6 Summary of Core DAX Measures

Table 6.3 summarizes the main DAX formulas created in Power BI for this project:

Table 6.3: Summary of Core DAX Measures.

Measure Name	**Home Table**	**DAX Formula / Logic**
Total Customers	FactCustomerRFMLive	`DISTINCTCOUNT(CustomerKey)`
Total Customer Spend	FactCustomerRFMLive	`SUM(Monetary)`
Inactive Customers (>100d)	FactCustomerRFMLive	`CALCULATE([Total Customers], Recency > 100)`
Repeat Customer Rate	FactCustomerRFMLive	`VAR R = CALCULATE([Total Customers], Frequency >= 2) RETURN DIVIDE(R, [Total Customers], 0)`
Average Order Value (AOV)	FactCustomerRFMLive	`DIVIDE([Total Customer Spend], SUM(Frequency), 0)`
Avg Spend per Customer	FactCustomerRFMLive	`DIVIDE([Total Customer Spend], [Total Customers], 0)`
Monthly Customer Spend	FactCustomerSnapshot	`SUM(MonetarySnapshot)`

Bibliography

[1] Claus O. Wilke. *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media, 2019. ISBN: 978-1492031086.